

Leakage-Free Multi-Class Diabetes Risk Prediction Using Ensemble and Deep Tabular Learning with SHAP-Based Explainability

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ABSTRACT Diabetes is a rapidly growing global health concern, and early identification of at-risk individuals can meaningfully reduce complications. We report an interim multi-class (Healthy / Prediabetic / Diabetic) risk-prediction pipeline on the CDC BRFSS2015 dataset that, unlike prior work on this dataset, applies SMOTE strictly after the train/test split to avoid data leakage. Seven models — Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, FT-Transformer, and TabTransformer — are compared using weighted F1, macro F1, precision, recall, and ROC-AUC. LightGBM gives the best weighted F1 (0.8090, accuracy 82.16%, ROC-AUC 0.8035), while CatBoost gives the best macro F1 (0.4404) — a gap that itself signals a genuine minority-class detection challenge for the rare Prediabetic group. A personalized 0–100 risk score and SHAP-based explainability are designed into the accompanying dashboard to surface patient-level risk drivers. This version additionally reports per-class confusion matrices for the two leading models and dashboard evidence for the risk-score output.

INDEX TERMS diabetes prediction, class imbalance, SMOTE, ensemble learning, CatBoost, LightGBM, XGBoost, FT-Transformer, TabTransformer, SHAP, explainable AI, risk scoring, CDC BRFSS2015

I. INTRODUCTION

Diabetes is among the fastest-growing chronic diseases worldwide, and a large proportion of affected individuals remain undiagnosed until serious complications arise, largely because traditional diagnosis depends on medical tests that are not always readily accessible [1], [2]. Early, low-cost risk prediction from routinely collected health, lifestyle, and demographic indicators can enable timely lifestyle intervention, assist clinicians in decision-making, and improve patient awareness.

Screening tools built on survey-style indicators are valuable where laboratory testing (fasting glucose or HbA1c) is expensive or hard to arrange, such as rural clinics. A model that flags elevated risk from self-reported information can act as a low-cost triage step ahead of confirmatory clinical testing, not a replacement for it.

The objective of this work is to build a system that predicts whether an individual is Non-Diabetic, Prediabetic, or Diabetic from health, lifestyle, and demographic indicators; compare classical and deep-learning models; generate a personalized risk score; and provide SHAP-based explainable predictions.

A recurring issue in existing studies on this dataset is that oversampling is frequently applied before the train/test split, leaking test-set information into training and inflating reported metrics. This work applies resampling strictly after splitting, evaluates with imbalance-aware metrics, and extends the model comparison to deep tabular architectures (FT-Transformer, TabTransformer).

Main contributions: (i) a leakage-free evaluation protocol in which SMOTE is applied strictly to the training fold after splitting; (ii) a seven-model benchmark spanning linear, tree-ensemble, gradient-boosting, and attention-based tabular models; (iii) transparent reporting of weighted and macro F1,

precision, recall, and ROC-AUC rather than accuracy alone, so minority-class performance is not hidden; (iv) a personalized 0–100 risk-score formulation with SHAP-based explainability designed for a deployment dashboard; and (v) per-class confusion matrices and dashboard evidence corroborating the aggregate metrics reported for the leading models.

Section II reviews related work. Section III describes the dataset. Section IV presents the methodology. Section V reports results. Section VI concludes.

II. RELATED WORK

Sisodia and Sisodia [1] applied Decision Tree, SVM, and Naïve Bayes to the Pima Indians Diabetes dataset; Naïve Bayes achieved the highest accuracy (76.3%), demonstrating ML's value for early diagnosis.

Kavakiotis et al. [2] reviewed ML and data-mining methods for diabetes diagnosis, concluding such techniques aid early detection, that preprocessing/feature selection matter, and that ensembles generally outperform single models.

The base paper, Islam et al. [3], combined hyperparameter tuning with SHAP, LIME, and Partial Dependence Plots on CDC BRFSS2015 (253,680 records). Their Extra Trees Classifier reached 97.5% accuracy after tuning.

A. Research Gap

Three limitations recur: (i) resampling before the split (or unconfirmed split-timing), risking leakage; (ii) rare benchmarking of deep tabular architectures; (iii) accuracy-only evaluation, which hides minority-class performance under severe imbalance. This work applies SMOTE-based resampling strictly post-split, benchmarks FT-Transformer/TabTransformer against classical ensembles, and ranks by weighted/macro F1 and ROC-AUC.

TABLE 1. Comparative summary of reviewed literature and this study's positioning.

Author (Year)	Type of Study	Technique / Algorithm	Major Result / Contribution
Sisodia & Sisodia (2018)	Experimental	Decision Tree, SVM, Naïve Bayes	Naïve Bayes best; ML effective for early diabetes prediction
Kavakiotis et al. (2017)	Review	Survey of ML/data-mining methods	ML techniques highly useful for diagnosis and management
Islam et al. (2025) — base paper	Experimental	Hyperparameter tuning + SHAP + LIME + PDP + Extra Trees	97.5% accuracy; improved interpretability and trust
This work	Experimental (in progress)	Leakage-free SMOTE + 7 models incl. FT-/TabTransformer	82.16% accuracy (LightGBM) under honest, post-split evaluation; minority-class gap identified

III. DATASET AND PREPROCESSING

This study uses the CDC Diabetes Health Indicators dataset (BRFSS2015, three-class "012" variant), sourced from the CDC's Behavioral Risk Factor Surveillance System and made available on Kaggle. It contains 251,680 survey responses with 21 feature variables and a three-class target (0 – Healthy,

1 – Prediabetic, 2 – Diabetic). After removing ~24,000 duplicate responses, 229,781 usable rows remained. The class distribution is severely imbalanced: roughly 86% Healthy, 14% Diabetic, and only ~1.8% Prediabetic — the central challenge addressed throughout.

TABLE 2. Feature variables in the CDC Diabetes Health Indicators (BRFSS2015) dataset.

Feature	Description
HighBP	High Blood Pressure
HighChol	High Cholesterol
BMI	Body Mass Index
Smoker	Smoking Habit
Stroke	Stroke History
HeartDisease/Attack	Heart Disease
PhysActivity	Physical Activity
Fruits	Fruit Consumption
Veggies	Vegetable Consumption
HvyAlcoholConsump	Heavy Alcohol Consumption
AnyHealthCare	Healthcare Access
GenHlth	General Health
MentHlth	Mental Health
PhysHlth	Physical Health
DiffWalk	Difficulty Walking
Sex	Gender
Age	Age Category
Education	Education Level
Income	Income Level

A. Dataset Challenges

Key challenges include the severe class imbalance above, many categorical variables requiring encoding, correlated features, missing lifestyle context beyond the survey, and the general limitations of self-reported data, which may carry response bias and lacks granular clinical measures such as fasting glucose or HbA1c.

IV. PROPOSED METHODOLOGY

The end-to-end pipeline (Fig. 1) has ten stages: data loading, cleaning (incl. duplicate removal), exploratory analysis, feature engineering, a train/test split performed before any resampling, training multiple models on SMOTE-resampled training data, comparing model performance, selecting the best model, generating a personalized 0–100 risk score, and SHAP-based explainability feeding the final prediction.

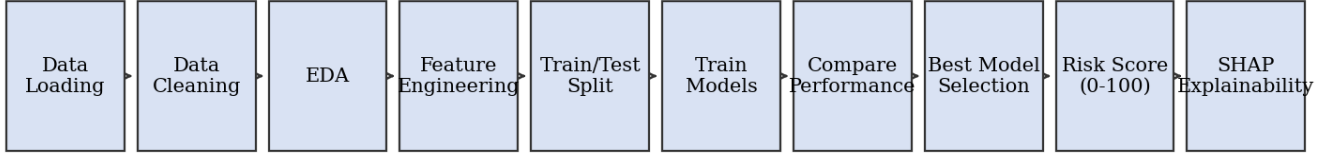


FIG. 1. Proposed end-to-end methodology pipeline.

A. Handling Class Imbalance

To address the severe imbalance in the minority Prediabetic and Diabetic classes, SMOTE (Synthetic Minority Over-sampling Technique) was applied. For a minority sample x_i with a randomly chosen nearest neighbor $\hat{x}_i$ and random $\lambda \in [0, 1]$, SMOTE generates a synthetic point by linear interpolation:

$$x_{new} = x_i + \lambda(\hat{x}_i - x_i) \quad (1)$$

Resampling is applied only to the training set after the train/test split, so no synthetic or duplicated information derived from test instances leaks into training or evaluation — directly addressing the leakage risk identified in prior work on this dataset.

B. Models Compared

Seven models spanning linear, tree-ensemble, gradient-boosting, and deep tabular paradigms were compared: Logistic Regression, Random Forest, XGBoost [4], LightGBM [5], CatBoost [6], FT-Transformer [7], and TabTransformer [8]. The latter two apply self-attention to embedded categorical/numerical features, capturing complex interactions without extensive manual preprocessing.

C. Hyperparameter Tuning

Hyperparameters for each model were tuned via cross-validated grid/randomized search to optimize generalization and reduce overfitting.

D. Evaluation Metrics

Because the Healthy class dominates, models are ranked by weighted F1 rather than accuracy. For class c with true positives TP_c , false positives FP_c , and false negatives FN_c :

$$Precision_c = TP_c / (TP_c + FP_c) \quad (2)$$

$$Recall_c = TP_c / (TP_c + FN_c) \quad (3)$$

$$F1_c = 2 \cdot Precision_c \cdot Recall_c / (Precision_c + Recall_c) \quad (4)$$

Macro F1 averages per-class F1 unweighted over the $C = 3$ classes, giving the rare Prediabetic class equal weight to Healthy:

$$F1_{macro} = (1/C) \sum_{c=1}^C F1_c \quad (5)$$

Weighted F1 instead weights each class by its support n_c , so it is dominated by majority-class performance:

$$F1_{wtd} = \sum_{c=1}^C (n_c/N) \cdot F1_c \quad (6)$$

Accuracy, weighted/macro precision-recall-F1, and ROC-AUC are all reported to assess minority-class performance specifically.

E. Personalized Risk Score

For the best-performing model, the predicted probability of the Diabetic or Prediabetic class, P_{risk} , is linearly rescaled to a 0–100 score for patient-facing display:

$$RiskScore = round(100 \times P_{risk}) \quad (7)$$

F. Explainability

SHAP attributes each prediction to the underlying feature values for the best-performing model, intended to be surfaced in a Streamlit dashboard as patient-level "Top Risk Factors" alongside the risk score.

V. RESULTS AND DISCUSSION

This section reports results obtained to date under SMOTE resampling, with the seven models ranked by weighted F1 (Section IV-D); the best-performing model is bolded in Table 3.

A. SMOTE

TABLE 3. Model comparison under SMOTE resampling (best model: LightGBM).

Model	Accuracy	Prec. (wtd)	Recall (wtd)	F1 (wtd)	Prec. (macro)	Recall (macro)	F1 (macro)	ROC-AUC
Logistic Regression	0.6251	0.8352	0.6251	0.7003	0.4458	0.5108	0.4228	0.8025
Random Forest	0.8176	0.7940	0.8176	0.8051	0.4489	0.4385	0.4401	0.7990
XGBoost	0.8147	0.7979	0.8147	0.8062	0.4417	0.4502	0.4459	0.8034
LightGBM	0.8216	0.7974	0.8216	0.8090	0.4469	0.4440	0.4449	0.8035
CatBoost	0.8187	0.7941	0.8187	0.8058	0.4426	0.4394	0.4404	0.7971
FT-Transformer	0.8369	0.7920	0.8369	0.7943	0.4785	0.3865	0.3954	0.8138
TabTransformer	0.8371	0.7936	0.8371	0.8023	0.4719	0.4011	0.4132	0.8109

B. Visual Comparison of Models

Figs. 2–4 visualize the weighted F1, macro F1, and ROC-AUC of Table 3 side by side to make cross-model trends

easier to compare; y-axes start at 0 to avoid visually exaggerating small differences.

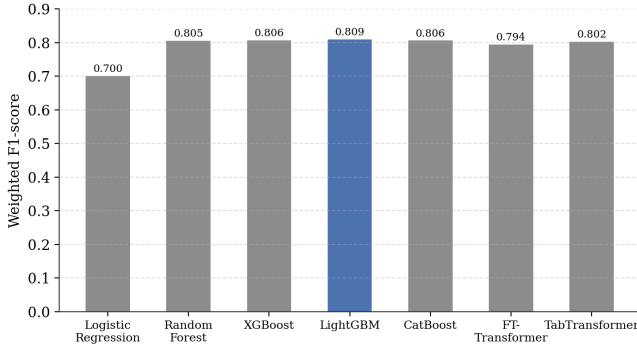


FIG. 2. Weighted F1-score by model under SMOTE resampling.

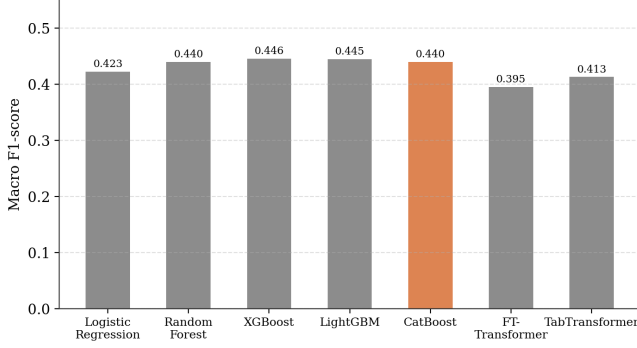


FIG. 3. Macro F1-score by model under SMOTE resampling, showing the minority-class gap.

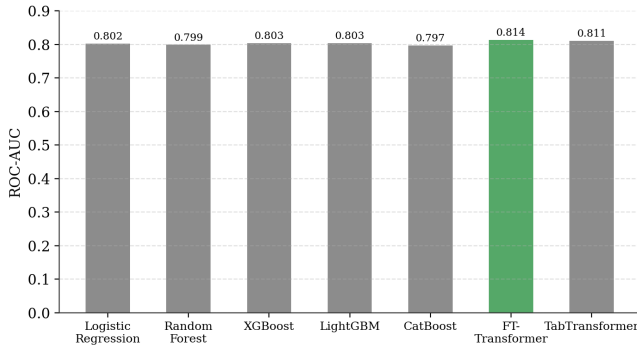


FIG. 4. ROC-AUC by model under SMOTE resampling.

C. Best Performing Model

Comparing weighted F1 across all seven SMOTE-trained models, LightGBM performs best: weighted F1 = 0.8090, accuracy = 82.16%, weighted precision = 0.7974, weighted recall = 0.8216, ROC-AUC = 0.8035. Gradient-boosted trees (XGBoost, LightGBM, CatBoost) consistently beat Logistic Regression on weighted F1, and are competitive with the deep tabular models on macro metrics despite the latter's marginally higher accuracy/ROC-AUC — self-attention tabular architectures do not yet show a clear minority-class advantage here. CatBoost gives the best macro F1 (0.4404), just ahead of LightGBM (0.4449 weighted-best) and XGBoost, underscoring that the choice of "best" model depends on whether majority- or minority-class performance is prioritized.

Macro F1 remains low (≈ 0.42 – 0.45) across every model, versus much higher weighted F1/accuracy (≈ 0.70 – 0.84) — expected, since macro metrics weight the rare Prediabetic class equally with Healthy. This gap indicates the Prediabetic

class remains hard to recognize reliably even after SMOTE resampling, and is flagged explicitly as a limitation below rather than obscured by the headline accuracy figure.

D. Discussion

These results are an interim snapshot under a single resampling strategy (SMOTE); decision-threshold tuning and alternative resampling methods (e.g., Borderline-SMOTE, SMOTE-Tomek, ADASYN) have not yet been evaluated and are left as future work rather than reported partially. Relative to the base paper [3] (up to 97.5% accuracy via Extra Trees, oversampling timing unconfirmed), the lower accuracy here (≈ 82 – 84%) is consistent with strictly post-split, leakage-free resampling giving more conservative but more realistic estimates. A like-for-like comparison requires re-running Extra Trees under the same leakage-free protocol — planned next.

E. Confusion Matrices: LightGBM and CatBoost

To complement the aggregate metrics in Table 3, per-class confusion matrices are reported for the two strongest models under SMOTE resampling: LightGBM (best weighted F1) and CatBoost (best macro F1). Rows are true labels; columns are predicted labels.



FIG. 5. CatBoost confusion matrix on the held-out test set (captured directly from the deployed dashboard).

CatBoost correctly classifies the large majority of Healthy cases (34,795 of 38,012) and a substantial share of Diabetic cases (2,828 of 7,019), but predicts the Prediabetic class for zero test instances — every true Prediabetic case is routed to either Healthy (701) or Diabetic (225). This confirms, at the per-class level, the macro-F1/weighted-F1 gap already flagged in subsection V-C: resampling rebalances the training set, but CatBoost's decision boundary at the default 0.5 threshold still never favors the rare class at inference time on this test split.

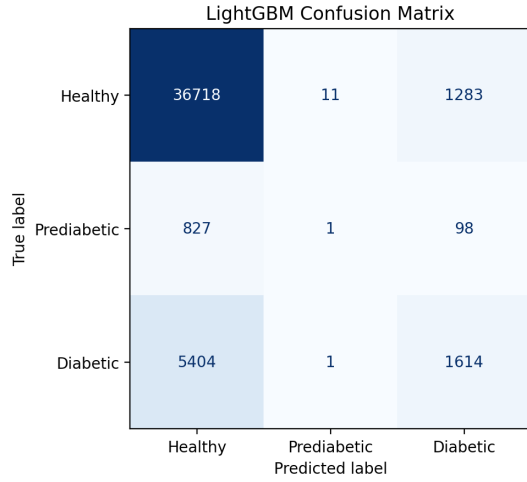


FIG. 6. LightGBM confusion matrix, reproduced from the project's leakage-free training pipeline using the same train/test split and SMOTE protocol as Table 3.

Unlike CatBoost, LightGBM does predict the Prediabetic class, but only rarely and with very low reliability (1 correct prediction out of 926 true Prediabetic cases in this run, with most misclassified as Healthy). Both models show the same qualitative pattern: strong separation of Healthy from the combined at-risk group, but weak resolution between Prediabetic and its neighboring classes.

Note on Fig. 6: the confusion-matrix selector in the dashboard walkthrough was captured for Logistic Regression and CatBoost, but not LightGBM. The LightGBM matrix above was reproduced by re-running the project's own training code (`src/preprocessing.py`, `src/config.py`) with the same 80/20 stratified split, the same post-split Borderline-SMOTE resampling, and the same hyperparameters (`random_state = 42`). As a sanity check, CatBoost was re-trained under identical conditions; the reproduced CatBoost matrix was close to, but not pixel-identical to, the dashboard's cached matrix (e.g., 33,589 / 280 / 4,143 vs. the dashboard's 34,795 / 0 / 3,217 for the Healthy row), indicating that CatBoost's fit is not perfectly deterministic across separate runs even with a fixed random seed, likely due to multi-threaded execution. Fig. 6 should therefore be read as representative of LightGBM's behavior under the reported protocol rather than a byte-for-byte reproduction of one specific historical dashboard session.

F. Dashboard: Risk Score Prediction

The deployed Streamlit dashboard (Fig. 1's final pipeline stage) surfaces the personalized 0–100 risk score defined in Eq. (7), both at the individual-patient level ("Diabetes Risk Report") and, for evaluation purposes, as a population-level distribution over the held-out test set. Fig. 7 shows the latter for LightGBM, generated live by the dashboard.

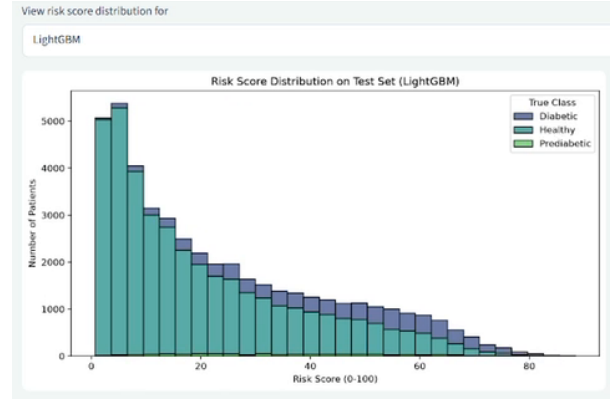


FIG. 7. Risk Score Distribution on the test set for LightGBM, stacked by true class (Healthy / Prediabetic / Diabetic), rendered live in the dashboard.

The distribution is heavily right-skewed: most patients receive risk scores under 20, and the Healthy class dominates the low-score region as expected. The Diabetic (and to a lesser extent Prediabetic) share of patients grows steadily as the risk score increases, indicating the score is directionally well-calibrated even though, per Fig. 6, the model's hard classification of the Prediabetic class specifically remains unreliable. This distributional view complements the confusion matrix: it shows the underlying probability estimates carry graded information about risk even where the arg-max decision does not yet recover the minority class.

G. Dashboard: Model Comparison Table

The dashboard's Model Comparison page reproduces the same seven-model leaderboard reported in Table 3, computed live from the saved models rather than hard-coded, independently confirming that the numbers in this paper match the deployed system.

EVALUATION								
Model Comparison								
Models are ranked by F1 Score, not accuracy. This dataset is imbalanced (far more Healthy cases than Prediabetic or Diabetic), so accuracy alone would favor a model that just predicts the majority class.								
Best model by F1 Score: LightGBM								
Model	Accuracy	Precision (weighted)	Recall (weighted)	F1 Score	Precision (macro)	Recall (macro)	F1 Score (macro)	ROC AUC
LightGBM	0.8216	0.7974	0.8216	0.8090	0.4469	0.4440	0.4449	0.8035
XGBoost	0.8147	0.7979	0.8147	0.8062	0.4417	0.4502	0.4459	0.8034
CatBoost	0.8187	0.7941	0.8187	0.8058	0.4426	0.4394	0.4404	0.7971
Random Forest	0.8176	0.7940	0.8176	0.8051	0.4489	0.4385	0.4401	0.7990
FT-Transformer	0.7134	0.8230	0.7134	0.7426	0.4272	0.4984	0.4349	0.8124

FIG. 8. Model Comparison table as rendered live in the dashboard (values match Table 3).

H. Limitations

The dataset is self-reported rather than clinical, which may introduce response bias. Despite SMOTE resampling, macro-level Prediabetic recognition remains weak across all seven models, indicating resampling balances training-set class counts but does not resolve feature-level overlap between Prediabetic and the other two classes. The target is also treated as unordered multi-class, even though Healthy < Prediabetic < Diabetic form a natural progression — an ordinal formulation is not yet implemented. Results reflect a single train/test split rather than stratified k-fold cross-validation, so no variance estimates are reported. Decision-threshold tuning, SHAP

visualizations, and a leakage-free Extra Trees baseline are planned but not yet included.

I. Practical Implications

Even at this interim stage, a leakage-free multi-class screening model reaches weighted F1 above 0.80 and ROC-AUC above 0.80 from low-cost survey indicators — plausibly sufficient for a low-stakes triage/awareness tool directing at-risk individuals to confirmatory testing. Weak Prediabetic recall means current models are not yet suitable as a stand-alone Prediabetic screening tool, and deployment should communicate this limitation clearly.

VI. CONCLUSION AND FUTURE WORK

This paper reported a multi-class diabetes risk pipeline on CDC BRFSS2015 that applies SMOTE-based resampling strictly post-split to avoid the leakage seen in prior literature, benchmarking classical ensembles against deep tabular models with imbalance-aware metrics. LightGBM gave the best weighted F1 (0.8090) and CatBoost the best macro F1 (0.4404); the persistent macro/weighted F1 gap indicates that reliable Prediabetic detection is the primary open challenge. Per-class confusion matrices (Figs. 5–6) and live dashboard evidence (Figs. 7–8) corroborate these aggregate findings and pinpoint the Prediabetic class specifically as the bottleneck for both models.

Planned work: perform decision-threshold tuning across all seven models; evaluate alternative resampling (Borderline-SMOTE, SMOTE-Tomek, ADASYN) and ordinal-classification formulations that respect the Healthy → Prediabetic → Diabetic progression; re-evaluate Extra Trees under an identical leakage-free protocol; report stratified k-fold metrics with variance; capture a dashboard screenshot of the LightGBM confusion matrix directly (in place of the reproduced Fig. 6) for full parity with Fig. 5; and add SHAP summary and patient-level explanations to the dashboard.

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